UNDEF

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Carrera Lic. en Ciberdefensa

Álgebra I

Profesor a cargo: Lic. Nelson Monzón

Profesora adjunta: Prof. Maela Lombardi

Año: 2025

Carga horaria: 64

SISTEMAS CON PARÁMETROS:

1) $\begin{cases} x_1+3x_2+5x_3=2 \\ -3x_1-8x_2-13x_3=-5 \\ -x_1-2x_2-3x_3=k \end{cases} \longrightarrow \left[\begin{array}{ccc|c} 1 & 3 & 5 & 2 \\ -3 & -8 & -13 & -5 \\ -1 & -2 & -3 & k \end{array}\right] \xrightarrow[F_3+F_1]{F_2+3F_1}$

$\longrightarrow \left[\begin{array}{ccc|c} 1 & 3 & 5 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & k+2 \end{array}\right] \xrightarrow{F_3-F_2} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & k+1 \end{array}\right]$

$$rg(A)=2, \quad rg(A|b)=\begin{cases} 2 & \text{si } k=-1 \\ 3 & \text{si } k\neq -1 \end{cases}$$

CASOS:

- si $k\neq -1$: $\Longrightarrow$ **S. I.** $\left(rg(A)=2<3=rg(A|b)\right)$
- si $k=-1$: $\Longrightarrow$ **S.C.I.** $\left(rg(A)=rg(A|b)=2<3=\#(\text{incóg.})\right)$

OBS: EN NINGÚN CASO EL SISTEMA RESULTA C.D.

2) $\begin{cases} x_1-6x_2+3x_3=1 \\ 4x_1-14x_2+10x_3=6 \\ 5x_1+kx_2+13x_3=0 \end{cases} \longrightarrow \left[\begin{array}{ccc|c} 1 & -6 & 3 & 1 \\ 4 & -14 & 10 & 6 \\ 5 & k & 13 & 0 \end{array}\right] \xrightarrow{F_3-F_1-F_2}$

$\longrightarrow \left[\begin{array}{ccc|c} 1 & -6 & 3 & 1 \\ 4 & -14 & 10 & 6 \\ 0 & k+20 & 0 & -7 \end{array}\right] \xrightarrow{\frac{1}{2}F_2} \left[\begin{array}{ccc|c} 1 & -6 & 3 & 1 \\ 2 & -7 & 5 & 3 \\ 0 & k+20 & 0 & -7 \end{array}\right] \xrightarrow{F_2-2F_1} \left[\begin{array}{ccc|c} 1 & -6 & 3 & 1 \\ 0 & 5 & -1 & 1 \\ 0 & k+10 & 0 & -7 \end{array}\right] \longrightarrow$

$\xrightarrow{F_3-4F_2} \left[\begin{array}{ccc|c} 1 & -6 & 3 & 1 \\ 0 & 5 & -1 & 1 \\ 0 & k & 4 & -11 \end{array}\right] \xrightarrow{F_3-\frac{k}{5}F_2} \left[\begin{array}{ccc|c} 1 & -6 & 3 & 1 \\ 0 & 5 & -1 & 1 \\ 0 & 0 & 4+\frac{k}{5} & -11-\frac{k}{5} \end{array}\right]$

$$rg(A)=\begin{cases} 2 & \text{si } k=-20 \\ 3 & \text{si } k\neq -20 \end{cases}$$

CASO 3:

- Se $k=-20$: $\left[\begin{array}{ccc|c} 1 & -6 & 3 & 1 \\ 0 & 5 & -1 & 1 \\ 0 & 0 & 0 & -15 \end{array}\right] \longrightarrow$ S. I. $\left(rg(A)=2<3=rg(A|b)\right)$.

- Se $k \neq -20$: S. C. D. $\left(rg(A)=rg(A|b)=3=\#(\text{incóg.})\right)$

3) $\begin{cases} x_1 + kx_2 = 3 \\ kx_1 + 4x_2 = 6 \end{cases} \longrightarrow \left[\begin{array}{cc|c} 1 & k & 3 \\ k & 4 & 6 \end{array}\right] \xrightarrow{F_2 - kF_1} \left[\begin{array}{cc|c} 1 & k & 3 \\ 0 & 4-k^2 & 6-3k \end{array}\right]$

$4-k^2=0 \iff (k=2 \text{ o } k=-2)$

$\therefore rg(A) = \begin{cases} 2 & \text{si } -2 \neq k \neq 2 \\ 1 & \text{si } k=2 \text{ o } k=-2 \end{cases}$

CASO 3:

- $k=-2$: $\left[\begin{array}{cc|c} 1 & -2 & 3 \\ 0 & 0 & 12 \end{array}\right] \longrightarrow$ S. I. $\left(rg(A)=1<2=rg(A|b)\right)$

- $k=2$: $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 0 \end{array}\right] \longrightarrow$ S. C. I. $\left(rg(A)=rg(A|b)=1<2=\#(\text{incóg.})\right)$

POR LO TANTO:

1) $-2 \neq k \neq 2 \Rightarrow$ S. C. D.
2) $k=2 \Rightarrow$ S. C. I.
3) $k=-2 \Rightarrow$ S. I.

4) $\begin{cases} x_1 + 2x_3 = 1 \\ kx_2 + 3x_3 = 2 \\ (k^2-1)x_3 = k+1 \end{cases} \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & k & 3 & 2 \\ 0 & 0 & k^2-1 & k+1 \end{array}\right]$

CASOS:

- $k=1$: $\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 2 \end{array}\right] \longrightarrow$ S. I.

$(rg(A) = 2 < 3 = rg(A|b))$

- $k=-1$: $\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -1 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right] \longrightarrow$ S.C.I.

$(rg(A) = rg(A|b) = 2 < 3 = \#(\text{INCÓG.}))$

- $k=0$: $\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & -1 & 1 \end{array}\right] \xrightarrow{F_2+3F_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & -1 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 5 \end{array}\right]$

$rg(A) = 2 < 3 = rg(A|b) \Rightarrow$ S. I.

- $k \notin \{0, -1, 1\}$ $rg(A) = rg(A|b) = 3 = \#(\text{INCÓG.})$

$\therefore$ S. C. D.

MÁS EJEMPLOS:

1) $\left\{\begin{array}{l} x_1 + 2x_2 + 2x_3 = 2 \\ x_2 + 2x_3 = -2 \\ x_1 + \alpha x_2 = \beta \end{array}\right. \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & 2 & -2 \\ 1 & \alpha & 0 & \beta \end{array}\right] \xrightarrow{F_3 - F_1}$

$\longrightarrow \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & 2 & -2 \\ 0 & \alpha-2 & -2 & \beta-2 \end{array}\right] \xrightarrow{F_3-(\alpha-2)} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & 2-2\alpha & 2\alpha+\beta-6 \end{array}\right]$

CASOS:

- $\alpha \neq 1$: $rg(A) = rg(A|b) = 3 = \#(\text{INCÓG.}) \Rightarrow$ S.C.D.
- $\alpha = 1$: $\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & \beta-4 \end{array}\right]$

(.) $\beta \neq 4 \Rightarrow rg(A) = 2 < 3 = rg(A|b) \Rightarrow$ S. I.

(..) $\beta = 4 \Rightarrow rg(A) = rg(A|b) = 2 < 3 = \#(\text{INCÓG.}) \Rightarrow$ S.C.I.

ENTONCES:

- $\alpha \neq 1, \beta \in \mathbb{R} \Rightarrow$ S.C.D.
- $\alpha = 1, \beta \neq 4 \Rightarrow$ S.I.
- $\alpha = 1, \beta = 4 \Rightarrow$ S.C.I.

2) SABIENDO QUE $(0,1,-1)$ ES SOLUCIÓN DEL SISTEMA

$$\begin{cases} x_1 + x_2 - x_3 = 2 \\ \beta x_2 - 3x_3 = 2 \\ \alpha x_1 - x_2 - 3x_3 = 2 \end{cases}$$

ANALIZAR LAS SOLUCIONES DEL MISMO PARA LOS DISTINTOS VALORES DE α Y β.

S/

$$(A|b) = \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & \beta & -3 & 2 \\ \alpha & -1 & -3 & 2 \end{array}\right]$$

REEMPLAZAMOS $(x_1, x_2, x_3) = (0,1,-1)$ Y OBTENEMOS

$$\left.\begin{cases} 1-(-1)=2 \\ \beta - 3(-1) = 2 \\ \alpha \cdot 0 - 1 - 3\cdot(-1) = 2 \end{cases}\right) \sim \left\{\begin{array}{l} 2 = 2 \ \checkmark \\ \beta = -1 \\ -1+3 = 2 \ \checkmark \end{array}\right) \Rightarrow \boxed{\beta = -1}$$

$$\therefore (A|b) = \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & -3 & 2 \\ \alpha & -1 & -3 & 2 \end{array}\right] \xrightarrow{F_3 - F_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & -3 & 2 \\ \alpha & 0 & 0 & 0 \end{array}\right] \xrightarrow{F_1 + F_2}$$

$$\longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & -4 & 4 \\ 0 & -1 & -3 & 2 \\ \alpha & 0 & 0 & 0 \end{array}\right] \xrightarrow{-F_2} \left[\begin{array}{ccc|c} 1 & 0 & -4 & 4 \\ 0 & 1 & 3 & -2 \\ \alpha & 0 & 0 & 0 \end{array}\right] \xrightarrow{F_3 - \alpha F_1} \left[\begin{array}{ccc|c} 1 & 0 & -4 & 4 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 4\alpha & -4\alpha \end{array}\right]$$

(CASOS):

- $\alpha \neq 0 \Rightarrow rg(A) = rg(A|b) = 3 = \#(\text{incóg}) \Rightarrow$ S.C.D.
- $\alpha = 0 \Rightarrow rg(A) = rg(A|b) = 2 < 3 = \#(\text{incóg}) \Rightarrow$ S.C.I.

RTA.:

DEBE SER $\beta = -1$.

PARA $\beta = -1$ LOS CASOS SON:

1) $\alpha \neq 0 \Rightarrow$ S.C.D.

2) $\alpha = 0 \Rightarrow$ S.C.I.